
Software Requirements Specification

for

MrDresser

Version 1.0 approved

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May 6, 2025

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This SRS defines the requirements for the MrDresser **Personalized Fashion Recommender**, which profiles men with smart clothing recommendations based on their physical appearance e.g. age, face, weight, skin tone and provides personalized dress suggestions depending on the weather, event, occasion and location. It aims to automate and personalize the process of outfit selection using machine learning and user profiling.

1.2 Document Conventions

Bold headings indicate major sections. Plain text highlights important references. Requirement IDs follow the format FR-xx for Functional Requirements and NFR-xx for Non-Functional Requirements.

1.3 Intended Audience and Reading Suggestions

- **Developers:** Use this document to implement features.
- **Testers:** Base your test cases on the functional requirements.
- **Users:** Refer to the overview for an understanding of product behavior.
- **Project Managers:** Track progress and validate system deliverables.

1.4 Product Scope

MrDresser will act as a virtual wardrobe consultant. It provides style suggestions based on the user's biometric and contextual data, enhancing the personal shopping experience and increasing user satisfaction. It is particularly targeted toward working professionals, students, and style-conscious individuals.

1.5 References

- IEEE 830-1998 – Recommended Practice for Software Requirements Specifications
- OpenWeatherMap API for weather integration
- Python OpenCV for face and color detection

2. Overall Description

2.1 Product Perspective

MrDresser is a standalone web/mobile application. It utilizes external APIs for real-time weather and internal AI algorithms for biometric and contextual analysis.

2.2 Product Functions

- Profile creation with user attributes
- Smart dress recommendation engine
- Face and color detection for matching colors
- Event-based suggestion engine e.g., formal vs. casual
- Weather-aware outfit recommendations
- Location and event-based filtering

2.3 User Classes and Characteristics

- **General Users:** Input their information to receive suggestions
- **Fashion Consultants:** Use advanced settings for clients
- **System Admins:** Manage database, content updates

2.4 Operating Environment

- **Platforms:** Android, Web Browsers
- **OS:** Android 10+, Windows 10+, macOS
- **Languages:** Python (backend), HTML/CSS, JavaScript (frontend)

2.5 Design and Implementation Constraints

- Must work with real-time APIs for weather
- Must support image processing libraries like OpenCV, TensorFlow Lite
- Must use secure authentication

2.6 User Documentation

- In-app tutorials

- PDF user manual
- Help/FAQ section
- Contact support

2.7 Assumptions and Dependencies

- Users have internet access
- Accurate data input from users
- Dependencies: Camera access, external APIs (weather, maps)

3. System Features

3.1 System Feature 1

User Profile and Personalization

3.1.1 Description and Priority

This feature allows users to create a personalized profile by entering or capturing biometric data such as age, gender, weight, face shape, and skin tone. This data is used to tailor clothing recommendations.

Priority: High

Benefit: 9

Penalty: 8

Cost: 4

Risk: 6

3.1.2 Stimulus/Response Sequences

Stimulus 1: User opens the profile page.

Response: System displays a form for entering personal data name, age, weight, etc.

Stimulus 2: User submits profile details.

Response: System stores data and updates recommendation logic.

Stimulus 3: User uploads or captures an image.

Response: System analyzes the image to determine face shape and skin tone.

3.1.3 Functional Requirements

REQ-1: The system shall allow users to enter and update their age, gender, weight, and profession.

REQ-2: The system shall enable users to upload a profile picture for automatic face shape and skin tone detection.

REQ-3: The system shall analyze the uploaded image using a face detection algorithm to extract face shape and color information.

REQ-4: The system shall store the user's profile data securely and associate it with their account.

REQ-5: If an invalid image format is uploaded, the system shall return an error message: "Unsupported image format. Please upload JPG or PNG."

REQ-6: The system allow the user to view and edit their profile at any time.

REQ-7: The system shall use the profile data to filter and prioritize outfit recommendations accordingly.

3.2 System Feature 2

Vendor Profile and Business Management

3.2.1 Description and Priority

Vendors can create and manage their business profiles, including uploading products/services.

Priority: High

Benefit: 9

Penalty: 8

Cost: 4

Risk: 6

3.2.2 Stimulus/Response Sequences

Stimulus 1: Vendor opens profile management page.

Response: System displays form for business name, category, contact info, and location.

Stimulus 2: Vendor submits business profile.

Response: System stores data and displays vendor dashboard.

Stimulus 3: Vendor updates profile or uploads products.

Response: System validates and reflects changes in real-time.

3.2.3 Functional Requirements

REQ-6: The system shall allow vendors to create and update business profiles.

REQ-7: The system shall allow vendors to upload product listings with name, description, price, and images.

REQ-8: The system shall validate input fields and image formats.

REQ-9: The system shall store vendor data securely in the database.

3.3 System Feature 3

Product/Service Search

3.3.1 Description and Priority

This feature enables users to search for nearby shops and filter services/products based on input.

Priority: High

Benefit: 10

Penalty: 9

Cost: 5

Risk: 4

3.3.2 Stimulus/Response Sequences

Stimulus 1: User enters keyword in search bar.

Response: System displays relevant vendors/products.

Stimulus 2: User filters results by category/distance.

Response: System narrows down and re-renders results.

Stimulus 3: User clicks on a vendor.

Response: System displays vendor profile and available items.

3.3.3 Functional Requirements

REQ-10: The system shall allow users to perform keyword-based searches.

REQ-11: The system shall support filtering by product category and vendor type.

REQ-12: The system shall sort results by proximity (if location access is granted).

REQ-13: The system shall handle empty result sets gracefully with appropriate messages.

3.4 System Feature 4:

Location-Based Services (Optional)]

3.4.1 Description and Priority

Enhances usability with map-based search and directions.

Priority: Medium

Benefit: 8
Penalty: 6
Cost: 6
Risk: 7

3.4.2 Stimulus/Response Sequences

Stimulus 1: User allows location access.

Response: System retrieves user coordinates.

Stimulus 2: User views map.

Response: System marks nearby vendors on map interface.

Stimulus 3: User requests directions.

Response: System displays route using Maps API.

3.4.3 Functional Requirements

REQ-14: The system shall optionally request and retrieve user location.

REQ-15: The system shall display vendors on a map interface.

REQ-16: The system shall use Google Maps API to generate route directions.

REQ-17: The system shall not store user location without consent.

3.5 System Feature 5

Admin Dashboard and Management

3.5.1 Description and Priority

Admin oversees user/vendor activity and approves listings.

Priority: High

Benefit: 9

Penalty: 7

Cost: 5

Risk: 4

3.5.2 Stimulus/Response Sequences

Stimulus 1: Admin logs in.

Response: System displays dashboard with pending approvals.

Stimulus 2: Admin reviews a vendor/shop.

Response: System updates approval status and reflects listing visibility.

Stimulus 3: Admin removes a reportable entry.

Response: System deletes or flags content.

3.5.3 Functional Requirements

REQ-18: The system shall provide an admin dashboard for user/vendor monitoring.

REQ-19: The system shall allow admins to approve or reject vendor listings.

REQ-20: The system shall support the removal of inappropriate or invalid content.

REQ-21: The system shall log admin activities for auditing.

4. External Interface Requirements

4.1 User Interfaces

<Describe the logical characteristics of each interface between the software product and the users. This may include sample screen images, any GUI standards or product family style guides that are to be followed, screen layout constraints, standard buttons and functions (e.g., help) that will appear on every screen, keyboard shortcuts, error message display standards, and so on. Define the software components for which a user interface is needed. Details of the user interface design should be documented in a separate user interface specification.>

4.2 Hardware Interfaces

In Progress

4.3 Software Interfaces

In Progress

4.4 Communications Interfaces

The MrDresser system requires several communication functions to support real-time data interaction and service delivery:

- **Web Browser Compatibility:** The application will operate through modern web browsers (Chrome, Firefox, Edge) with responsive design support.
- **API Communications:** The system interacts with external services using standard HTTP/HTTPS protocols:
 - **Weather API:** Provides real-time weather data to personalize dress recommendations.

- **Maps API:** Supplies map-based data for vendor location and direction services.
- **Authentication Protocols:** All user roles (General User, Vendor, Consultant, Admin) will authenticate via secure HTTP(S) using token-based sessions or OAuth.
- **Message Formatting:** Data exchange between client and server will be in **JSON format** to ensure lightweight and structured communication.
- **Security & Encryption:**
 - All communication will be encrypted using **SSL/TLS protocols**.
 - Sensitive user data (e.g., profile info) will be transmitted and stored securely.
- **Data Transfer:** System is optimized for lightweight communication ensuring minimal latency; typical transfer rates depend on the client's internet speed.
- **Synchronization:** Real-time syncing ensures recommendation results and product updates reflect instantly using AJAX/RESTful APIs.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- NFR-01: Initial page load < 3 seconds
- NFR-02: Search results must return within 2 seconds

5.2 Safety Requirements

- Data validation on all inputs to prevent errors and misuse
- Backup strategy for MongoDB database

5.3 Security Requirements

- NFR-03: All passwords stored using hashing (e.g., bcrypt)
- NFR-04: Use HTTPS for all communications
- NFR-05: Admin access protected with strong authentication

5.4 Software Quality Attributes

- Maintainable: Modular code and reusable components
- Scalable: Support for increased number of users/shops
- Usable: Intuitive UI with minimal learning curve

5.5 Business Rules

- Only verified vendors can publish listings
- Admin approval is required before shop is visible
- Shops must update profiles periodically to remain listed

6. Other Requirements

- Optional multilingual support
- Compliance with local e-commerce regulations (if applicable)

Appendix A: Glossary

- **Vendor:** A shop owner or service provider using NearBuy
- **User:** A general user searching for products/services
- **MongoDB:** NoSQL database used for storing application data
- **React.js:** JavaScript frontend framework
- **API:** Application Programming Interface

Appendix B: Analysis Models

6.1 Use Case diagram:

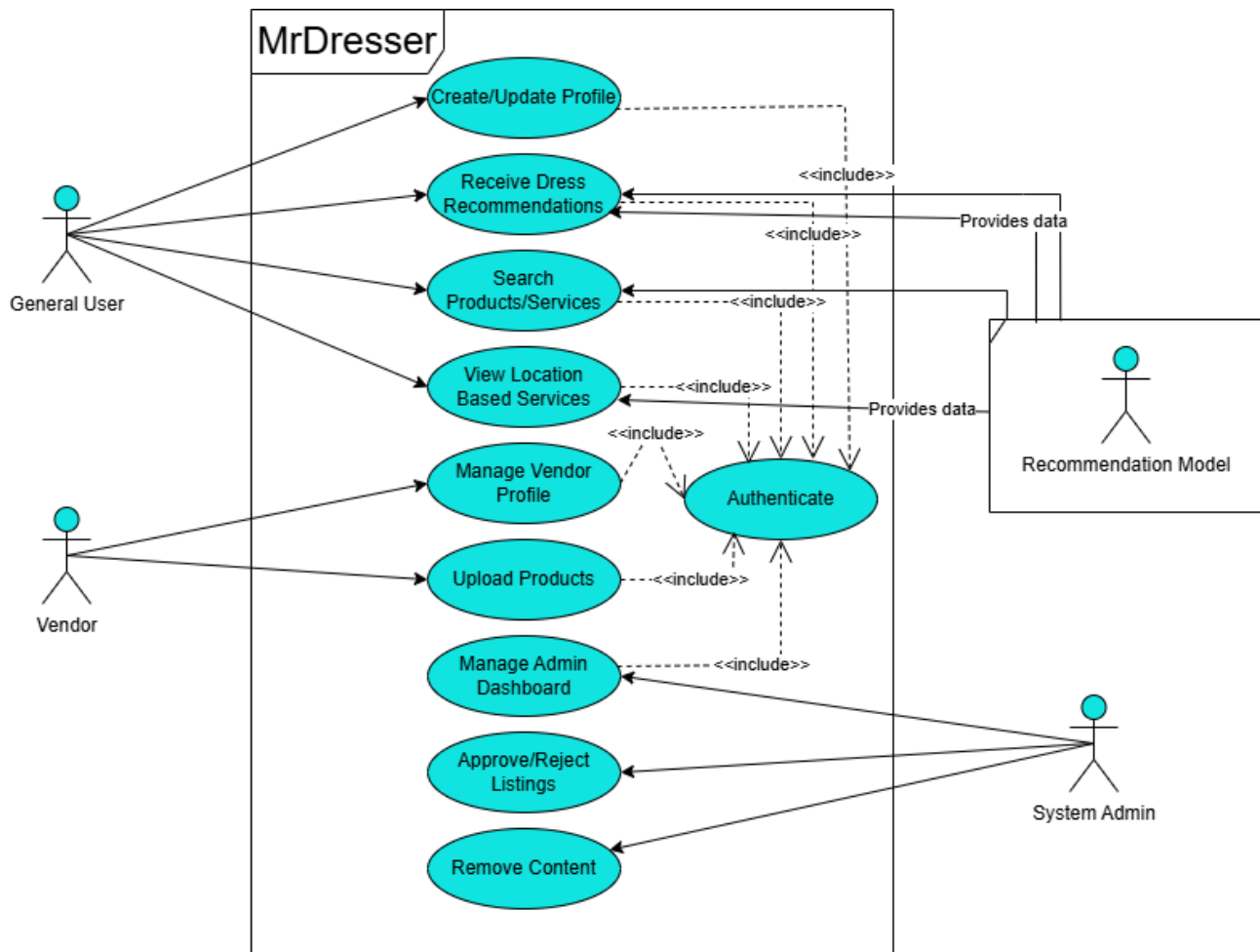


Figure 1

6.2 Level 0 DFD:

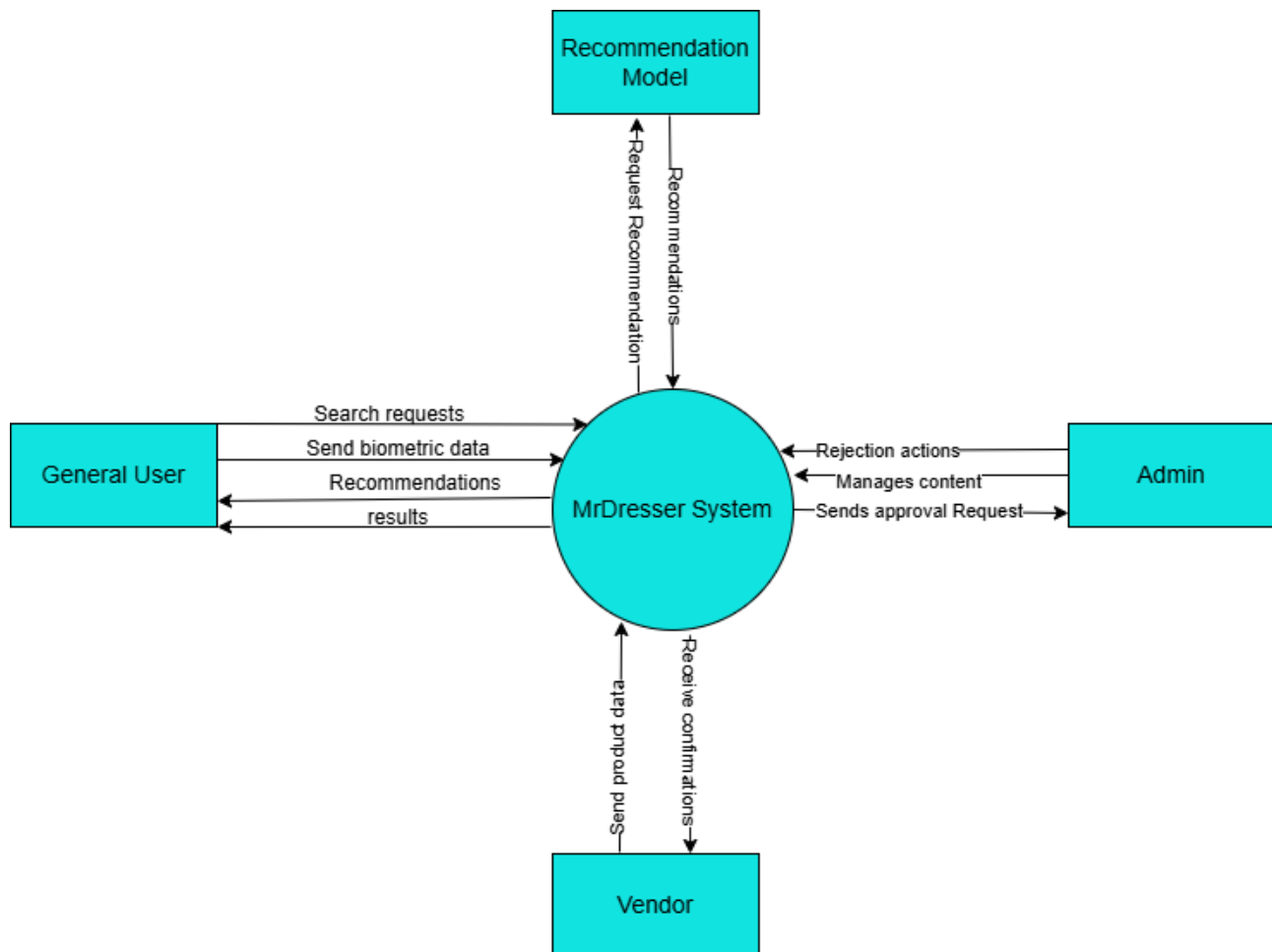


Figure 2

Appendix C: To Be Determined List

- TBD-1: Final decision on using Google Maps API
- TBD-2: Admin dashboard design layout